before the first and second laws as it provides a logical basis for the concept of temperature of the system.

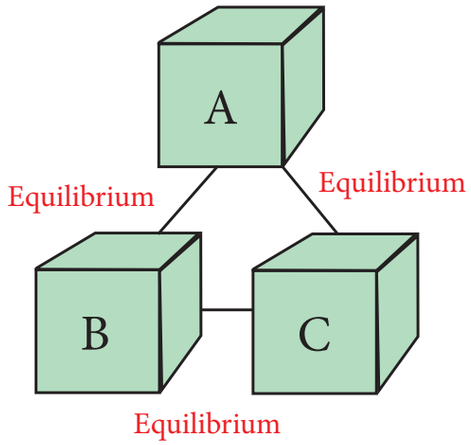

Figure : 7.5 Zeroth law of thermodynamics:

The law states that `If two systems are separately in thermal equilibrium with a third one, then they tend to be in thermal equilibrium with themselves'.

According to this law, if systems B and C separately are in thermal equilibrium with another system A, then systems B and C will also be in thermal equilibrium with each other. This is also the principle by which thermometers are used.

7.4 First Law of Thermodynamics:

The first law of thermodynamics, known as the law of conservation of energy, states that the total energy of an isolated system remains constant though it may change from one form to another.

When a system moves from state 1 to state 2, its internal energy changes from U_1 to U_2. Then change in internal energy

$$\Delta U = U_2 - U_1.$$

This internal energy change is brought about by the either absorption or evolution of heat and/or by work being done by/on the system.

Because the total energy of the system must remain constant, we can write the mathematical statement of the First Law as:

$$\Delta U = q + w \quad \text{----------(7.7)}$$

Where q - the amount of heat supplied to the system; w - work done on the system

Other statements of first law of thermodynamics

(1) Whenever an energy of a particular type disappears, an equivalent amount of another type must be produced.

(2) The total energy of a system and surrounding remains constant (or conserved)

(3) "Energy can neither be created nor destroyed, but may be converted from one form to another".

(4) "The change in the internal energy of a closed system is equal to the energy that passes through its boundary as heat or work".

(5) "Heat and work are two ways of changing a system's internal energy".

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